



IUS
INSTITUT
UNIVERSITAIRE
DES SCIENCES

Faculté des Sciences de Technologies

td9 Réseau1

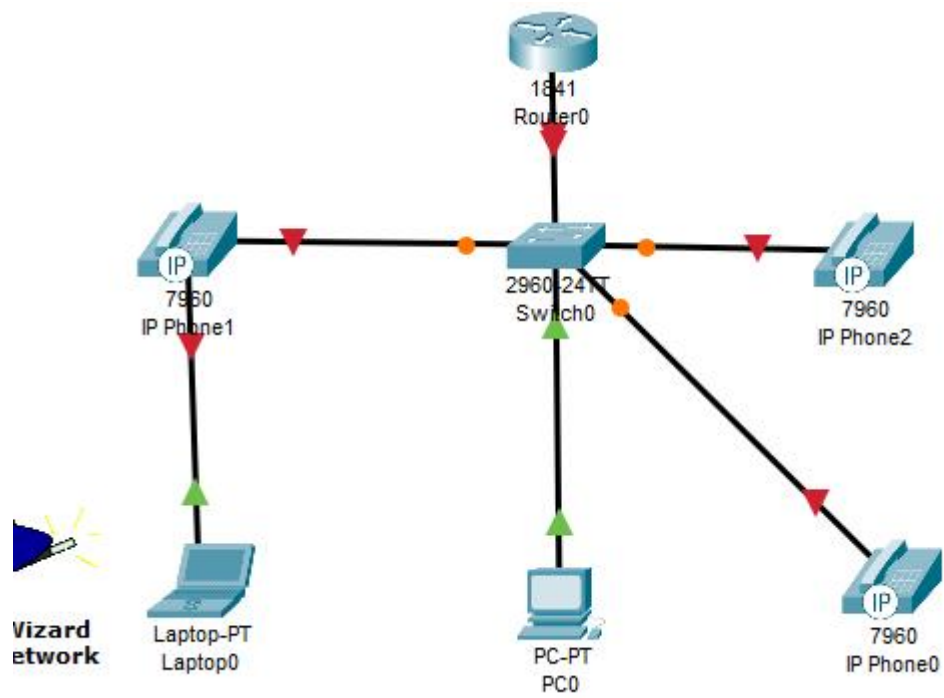
Nom : PIERRE

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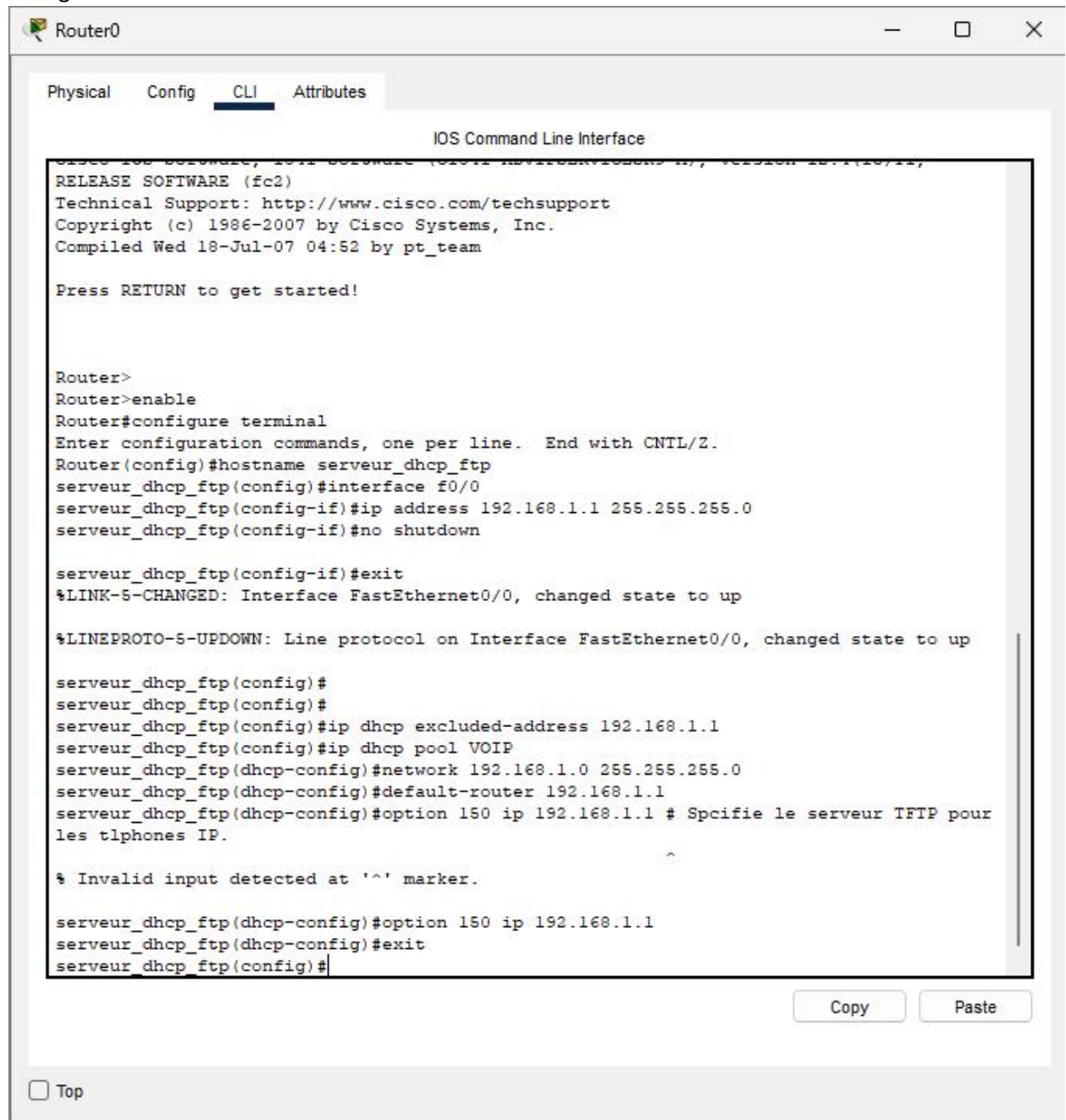
Niveau : L3-Sciences Informatiques

Professeur : Ismaël Saint Amour

1. Reproduisez cette topologie en configurant le VoIP.



configuration de la VoIP :



The screenshot shows a Cisco Router CLI window titled "Router0". The window has tabs for "Physical", "Config", "CLI", and "Attributes", with "CLI" selected. The main area displays the "IOS Command Line Interface". The text in the window is as follows:

```
RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Wed 18-Jul-07 04:52 by pt_team

Press RETURN to get started!

Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname serveur_dhcp_ftp
serveur_dhcp_ftp(config)#interface f0/0
serveur_dhcp_ftp(config-if)#ip address 192.168.1.1 255.255.255.0
serveur_dhcp_ftp(config-if)#no shutdown

serveur_dhcp_ftp(config-if)#exit
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

serveur_dhcp_ftp(config)#
serveur_dhcp_ftp(config)#
serveur_dhcp_ftp(config)#ip dhcp excluded-address 192.168.1.1
serveur_dhcp_ftp(config)#ip dhcp pool VOIP
serveur_dhcp_ftp(dhcp-config)#network 192.168.1.0 255.255.255.0
serveur_dhcp_ftp(dhcp-config)#default-router 192.168.1.1
serveur_dhcp_ftp(dhcp-config)#option 150 ip 192.168.1.1 # Spcifie le serveur TFTP pour
les tlphones IP.

% Invalid input detected at '^' marker.

serveur_dhcp_ftp(dhcp-config)#option 150 ip 192.168.1.1
serveur_dhcp_ftp(dhcp-config)#exit
serveur_dhcp_ftp(config)#
```

At the bottom of the window, there are "Copy" and "Paste" buttons, and a "Top" button with a checkbox.

```
serveur_dhcp_ftp(config)#ip source-address 192.168.1.1 port 2000
^
% Invalid input detected at '^' marker.

serveur_dhcp_ftp(config)#auto assign 1 to 5
^
% Invalid input detected at '^' marker.

serveur_dhcp_ftp(config)#exit
serveur_dhcp_ftp#
%SYS-5-CONFIG_I: Configured from console by console

serveur_dhcp_ftp# ?
Exec commands:
<1-99>      Session number to resume
auto        Exec level Automation
clear       Reset functions
clock       Manage the system clock
configure   Enter configuration mode
connect     Open a terminal connection
copy        Copy from one file to another
debug       Debugging functions (see also 'undebug')
delete      Delete a file
dir         List files on a filesystem
disable     Turn off privileged commands
disconnect  Disconnect an existing network connection
enable      Turn on privileged commands
erase       Erase a filesystem
exit        Exit from the EXEC
logout      Exit from the EXEC
mkdir       Create new directory
more        Display the contents of a file
no          Disable debugging informations
ping        Send echo messages
reload      Halt and perform a cold restart

serveur_dhcp_ftp#
```

3.2 Configurer les numéros de téléphone (ephone-dn) :

```

serveur_dhcp_ftp>enable
serveur_dhcp_ftp#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
serveur_dhcp_ftp(config)#ephone-dn 1
^
% Invalid input detected at '^' marker.

serveur_dhcp_ftp(config)#number 101
^
% Invalid input detected at '^' marker.

serveur_dhcp_ftp(config)#exit
serveur_dhcp_ftp#ephone-dn 2
^
% Invalid input detected at '^' marker.

serveur_dhcp_ftp#number 102
^
% Invalid input detected at '^' marker.

serveur_dhcp_ftp#exit
%SYS-5-CONFIG_I: Configured from console by console

```

QOS :

```

serveur_dhcp_ftp>enable
serveur_dhcp_ftp#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
serveur_dhcp_ftp(config)#class-map match-any VOICE
serveur_dhcp_ftp(config-cmap)# match protocol rtp
serveur_dhcp_ftp(config-cmap)# exit
serveur_dhcp_ftp(config)#policy-map VOICE_POLICY
serveur_dhcp_ftp(config-pmap)# class VOICE
serveur_dhcp_ftp(config-pmap-c)# priority percent 70
serveur_dhcp_ftp(config-pmap-c)# exit
serveur_dhcp_ftp(config-pmap)#interface f0/0
serveur_dhcp_ftp(config-if)# service-policy output VOICE_POLICY
serveur_dhcp_ftp(config-if)# exit
serveur_dhcp_ftp(config)#exit
serveur_dhcp_ftp#
%SYS-5-CONFIG_I: Configured from console by console

serveur_dhcp_ftp#

```

Copy

Paste

Configuration du commutateur

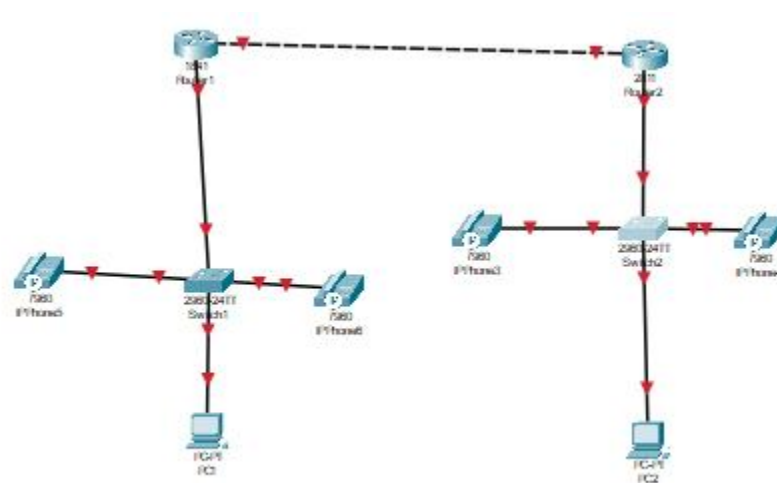
```

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname S1
S1(config)#interface range fa 0/1-24
S1(config-if-range)#switchport mode access
S1(config-if-range)#switchport voice vlan 1
S1(config-if-range)#exit
S1(config)#

```

Copy

2. Reproduire cette topologie



3.

Configuration appareil :

+> Routeur 1

Physical

Config

CLI

Attributes

IOS Command Line Interface

```
serveur_dhcp_ftp>enable
serveur_dhcp_ftp#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
serveur_dhcp_ftp(config)#hostname serveur_dhcp_ftp
serveur_dhcp_ftp(config)#interface f0/0
serveur_dhcp_ftp(config-if)#ip address 192.168.1.1 255.255.255.0
serveur_dhcp_ftp(config-if)#no shutdown
serveur_dhcp_ftp(config-if)#exit
serveur_dhcp_ftp(config)#interface f0/1
serveur_dhcp_ftp(config-if)#ip address 10.10.10.1 255.255.255.252
serveur_dhcp_ftp(config-if)#no shutdown
serveur_dhcp_ftp(config-if)#exit
serveur_dhcp_ftp(config)#ip dhcp pool VOIP
serveur_dhcp_ftp(dhcp-config)#network 192.168.1.0 255.255.255.0
serveur_dhcp_ftp(dhcp-config)#default-router 192.168.1.1
serveur_dhcp_ftp(dhcp-config)#option 150 ip 192.168.1.1
serveur_dhcp_ftp(dhcp-config)#exit
serveur_dhcp_ftp(config)#telephony-service
^
% Invalid input detected at '^' marker.

serveur_dhcp_ftp(config)#max-ephone 5
^
% Invalid input detected at '^' marker.

serveur_dhcp_ftp(config)#max-dn 5
^
% Invalid input detected at '^' marker.

serveur_dhcp_ftp(config)#ip source-address 192.168.1.1 port 2000
^
% Invalid input detected at '^' marker.

serveur_dhcp_ftp(config)#auto assign 1 to 5
^
% Invalid input detected at '^' marker.
```

```
serveur_dhcp_ftp>enable
serveur_dhcp_ftp#conf t
Enter configuration commands, one per line. End with CNTL/Z.
serveur_dhcp_ftp(config)#interface f0/1
serveur_dhcp_ftp(config-if)#router ospf 10
serveur_dhcp_ftp(config-router)#network 192.168.1.0 0.0.0.255 area 0
serveur_dhcp_ftp(config-router)#network 10.10.10.0 0.0.0.3 area 0
serveur_dhcp_ftp(config-router)#do wr
Building configuration...
[OK]
serveur_dhcp_ftp(config-router)#
```

Copy

Paste

+> Routeur 2

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname serveur_dhcp_ftp
serveur_dhcp_ftp(config)#interface f0/0
serveur_dhcp_ftp(config-if)#ip address 192.168.2.1 255.255.255.0
serveur_dhcp_ftp(config-if)#no shutdown

serveur_dhcp_ftp(config-if)#exit
serveur_dhcp_ftp(config)#interface f0/1
serveur_dhcp_ftp(config-if)#ip address 10.10.10.2 255.255.255.252
serveur_dhcp_ftp(config-if)#no shutdown

serveur_dhcp_ftp(config-if)#exit
serveur_dhcp_ftp(config)#ip dhcp pool VOIP
serveur_dhcp_ftp(dhcp-config)#network 192.168.2.0 255.255.255.0
serveur_dhcp_ftp(dhcp-config)#default-router 192.168.2.1
serveur_dhcp_ftp(dhcp-config)#option 150 ip 192.168.2.1
serveur_dhcp_ftp(dhcp-config)#exit
serveur_dhcp_ftp(config)#
serveur_dhcp_ftp(config)#Configuration des commutateurs
^
% Invalid input detected at '^' marker.

serveur_dhcp_ftp(config)#VLAN VoIP
^
% Invalid input detected at '^' marker.

serveur_dhcp_ftp(config)#S1 et S2
^
% Invalid input detected at '^' marker.

serveur_dhcp_ftp(config)#
serveur_dhcp_ftp(config)#R1
^
% Invalid input detected at '^' marker.
```

```
serveur_dhcp_ftp>en
serveur_dhcp_ftp#conf t
Enter configuration commands, one per line. End with CNTL/Z.
serveur_dhcp_ftp(config)#interface f0/1
serveur_dhcp_ftp(config-if)#router ospf 10
serveur_dhcp_ftp(config-router)#network 192.168.2.0 0.0.0.255 area 0
^
% Invalid input detected at '^' marker.

serveur_dhcp_ftp(config-router)#network 10.10.10.0 0.0.0.3 area 0
serveur_dhcp_ftp(config-router)#do wr
Building configuration...
[OK]
serveur_dhcp_ftp(config-router)#
00:17:22: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.1.1 on FastEthernet0/1 from LOADING
```


Physical Config CLI Attributes

IOS Command Line Interface

```
[OK]
serveur_dhcp_ftp(config-router)#
00:17:22: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.2.1 on FastEthernet0/1 from LOADING to
FULL, Loading Done

serveur_dhcp_ftp(config-router)#dial-peer voice 10 voip
^
% Invalid input detected at '^' marker.

serveur_dhcp_ftp(config-router)#destination-pattern 2..
^
% Invalid input detected at '^' marker.

serveur_dhcp_ftp(config-router)#session target ipv4:10.10.10.2
^
% Invalid input detected at '^' marker.

serveur_dhcp_ftp(config-router)#do wr
Building configuration...
[OK]
serveur_dhcp_ftp(config-router)#exit
serveur_dhcp_ftp(config)#dial-peer voice 10 voip
^
% Invalid input detected at '^' marker.

serveur_dhcp_ftp(config)#destination-pattern 2..
^
% Invalid input detected at '^' marker.

serveur_dhcp_ftp(config)#session target ipv4:10.10.10.2
^
% Invalid input detected at '^' marker.

serveur_dhcp_ftp(config)#do wr
Building configuration...
[OK]
serveur_dhcp_ftp(config)#exit
```

FULL, Loading Done

```
serveur_dhcp_ftp(config-router)#dial-peer voice 10 voip
^
% Invalid input detected at '^' marker.

serveur_dhcp_ftp(config-router)#destination-pattern 1..
^
% Invalid input detected at '^' marker.

serveur_dhcp_ftp(config-router)#session target ipv4:10.10.10.1
^
% Invalid input detected at '^' marker.

serveur_dhcp_ftp(config-router)#do wr
Building configuration...
[OK]
serveur_dhcp_ftp(config-router)#
```

Conv

Data

Copy

Paste

+> Switch 1,2

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname S1
S1(config)#interface range fa 0/1-24
S1(config-if-range)#switchport mode access
S1(config-if-range)#switchport voice vlan 1
S1(config-if-range)#exit
S1(config)#
```

3. Quel est le rôle de l'option 150 dans la configuration DHCP pour VoIP ?
Rep : L'option 150 dans la configuration DHCP fournit l'adresse IP du serveur TFTP aux téléphones VoIP. Cela permet aux téléphones de télécharger leurs fichiers de configuration nécessaires pour l'enregistrement et la configuration automatique sur le réseau VoIP, garantissant une intégration efficace avec le système.
4. Expliquez comment le Call Manager Express gère les appels entre les téléphones.
Rep : Le Call Manager Express (CME) gère les appels entre les téléphones en agissant comme un contrôleur SIP ou SCCP. Il assure l'enregistrement des téléphones IP, la gestion des numéros d'extension et l'établissement des appels en dirigeant les flux média directement entre les appareils, tout en optimisant les ressources réseau pour garantir une communication fluide.